

measuring and experimenting

digitalisierung und programmierung · prof. dr. nicolas meseth

- 1 measuring settles it
- 2 know what you measure
- 3 one reading is a guess
- 4 build evidence

part 1

measuring settles it

which one
lands first?

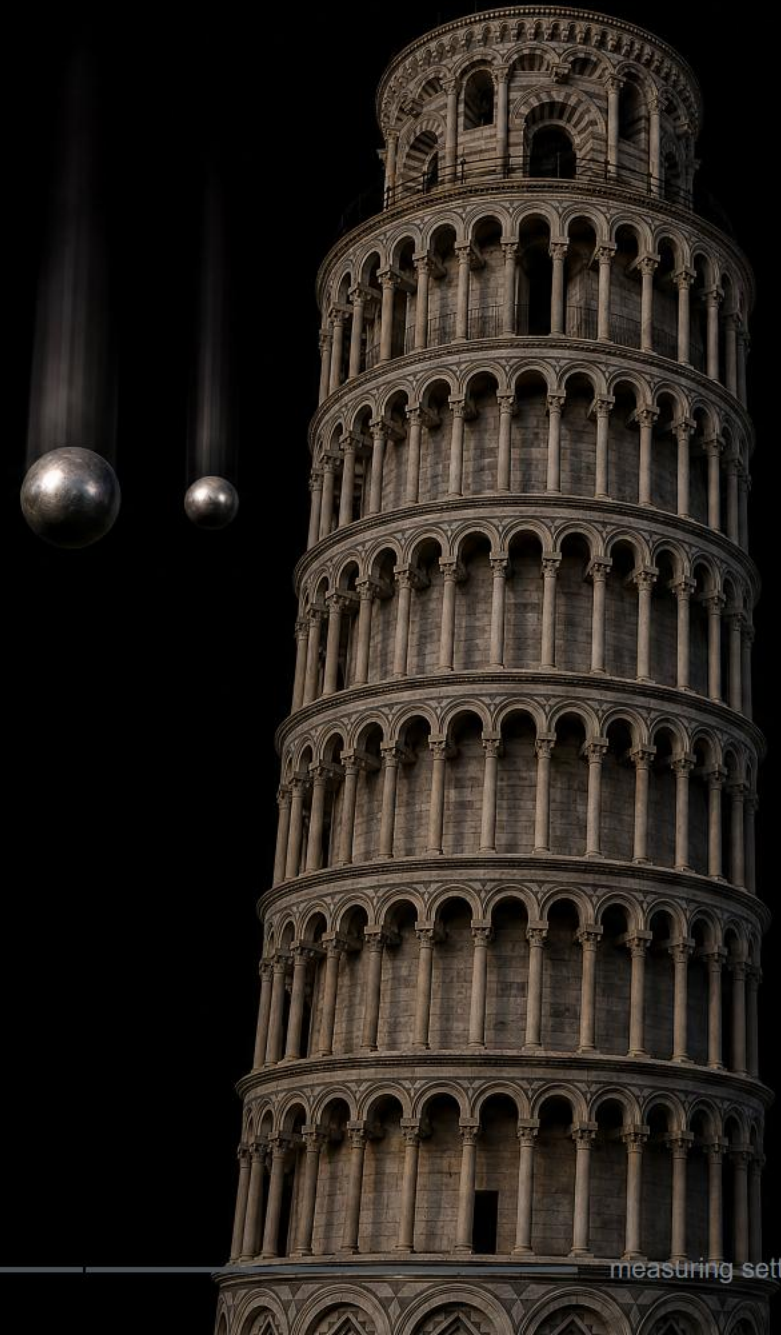


image: ai-generated (gpt-image-2)

measuring settles it

three ways to answer the same question

it sounds right

longer readings
collect more light

three ways to answer the same question

it sounds right

longer readings
collect more light

the assistant predicts it

double the
integration time

measuring settles it

three ways to answer the same question

it sounds right

longer readings
collect more light

the assistant predicts it

double the
integration time

the series shows it

41 / 50 → 47 / 50
recognised correctly



evidence from your setup

an explanation is a claim. a measurement can decide.

turn a claim into a test

question

does a longer window
improve recognition?

turn a claim into a test

question

prediction

does a longer window
improve recognition?



recognition should
improve

turn a claim into a test

question

prediction

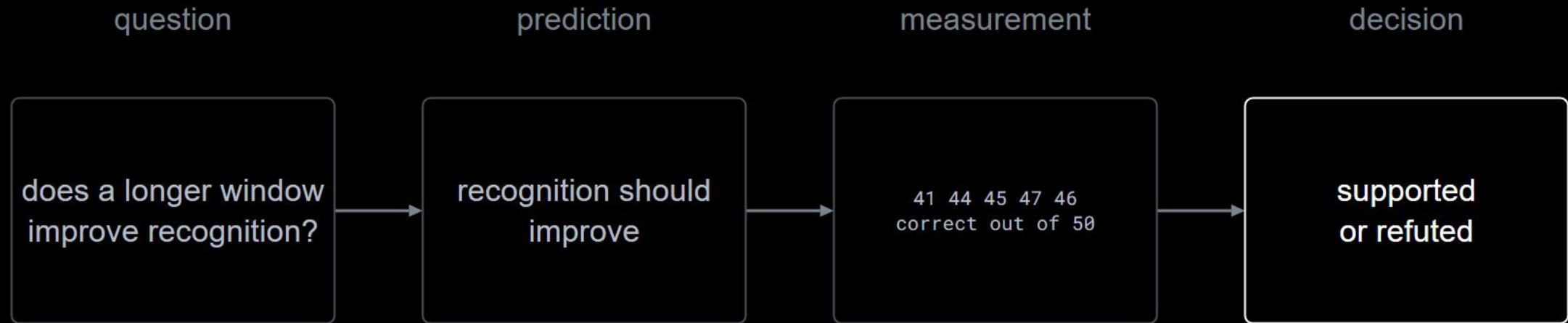
measurement

does a longer window
improve recognition?

recognition should
improve

41 44 45 47 46
correct out of 50

turn a claim into a test



write the expectation before you measure.

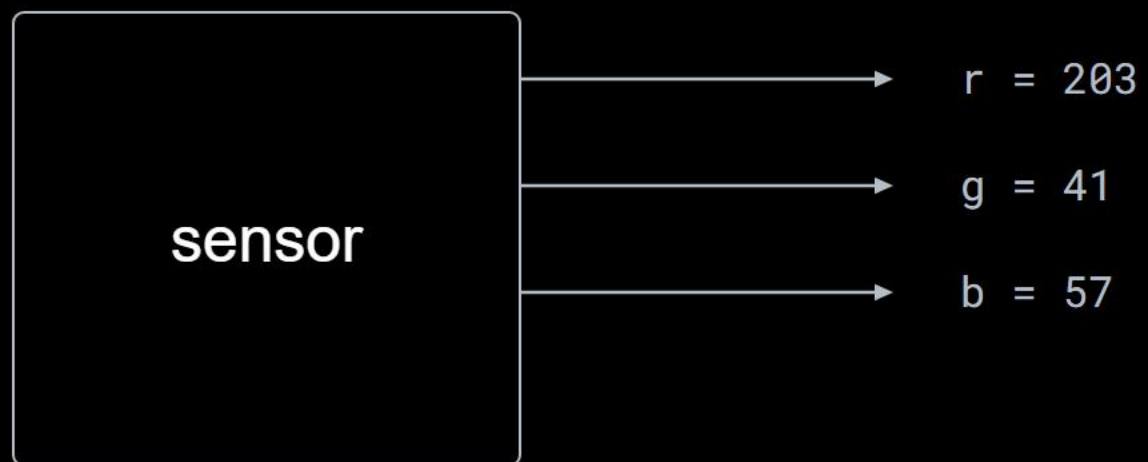
part 2

know what you measure

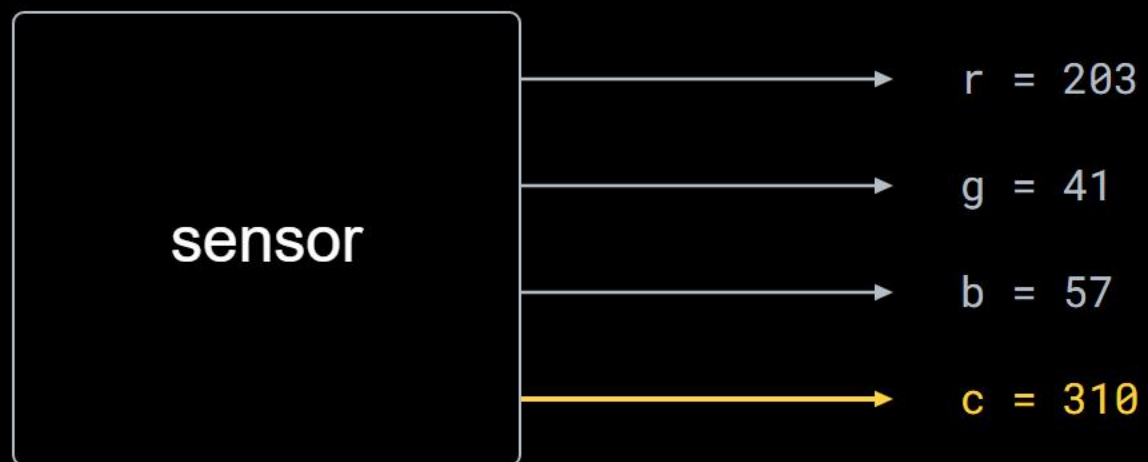
today we live at the input



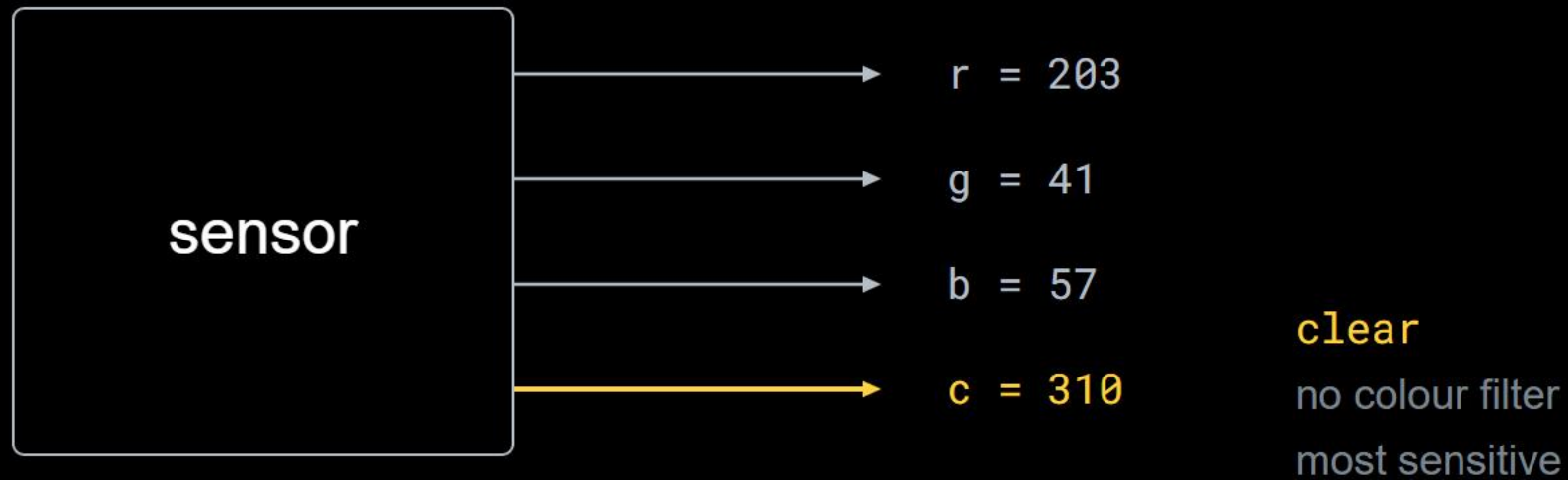
the sensor returns four numbers



the sensor returns four numbers



the sensor returns four numbers



your eyes see a colour. the program sees four values.

sent is not measured

what you send

$r = 255$
 $g = 0$
 $b = 0$



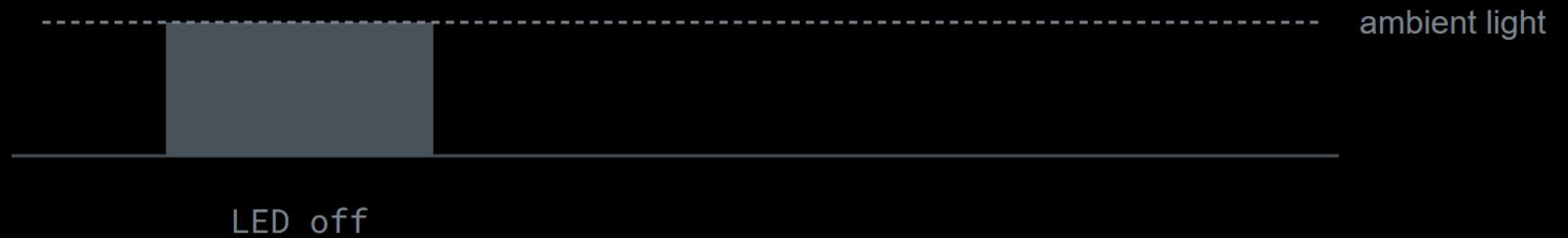
what arrives

$r = 203$
 $g = 41$
 $b = 57$
 $c = 310$

filters overlap · distance matters · the room measures with you

not broken. normal.

measure the dark first



measure the dark first



measure the dark first



measure the dark first



part 3

one reading is a guess

which value is true?

198

205

201

195

206

which value is true?

198

205

201

195

206

same LED · same settings · five readings

which value is true?

198

205

201

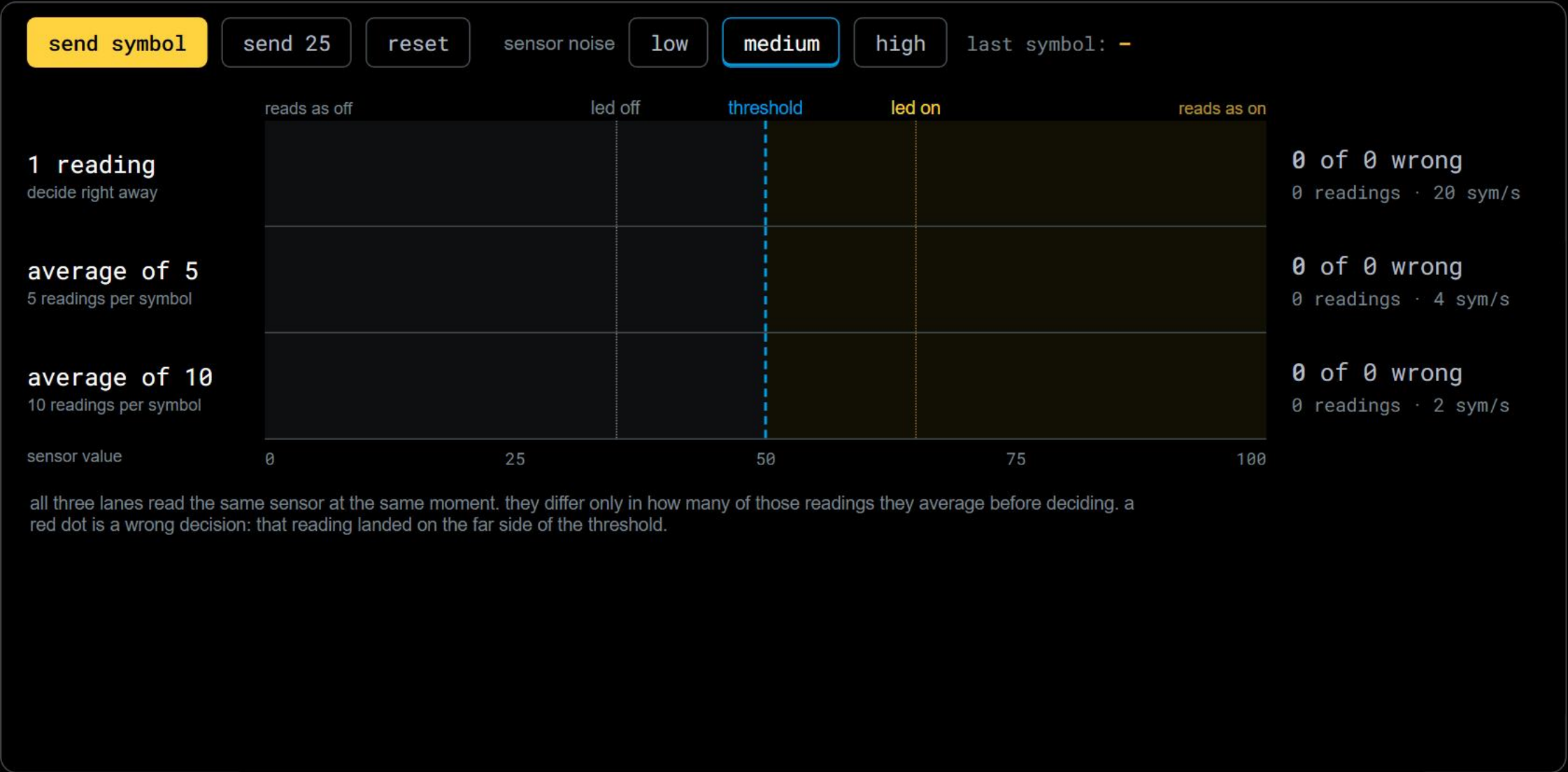
195

206

same LED · same settings · five readings

all of them are measurements

try it: how many readings per decision?



a series reveals what one reading hides

individual readings



averages of five



a series reveals what one reading hides

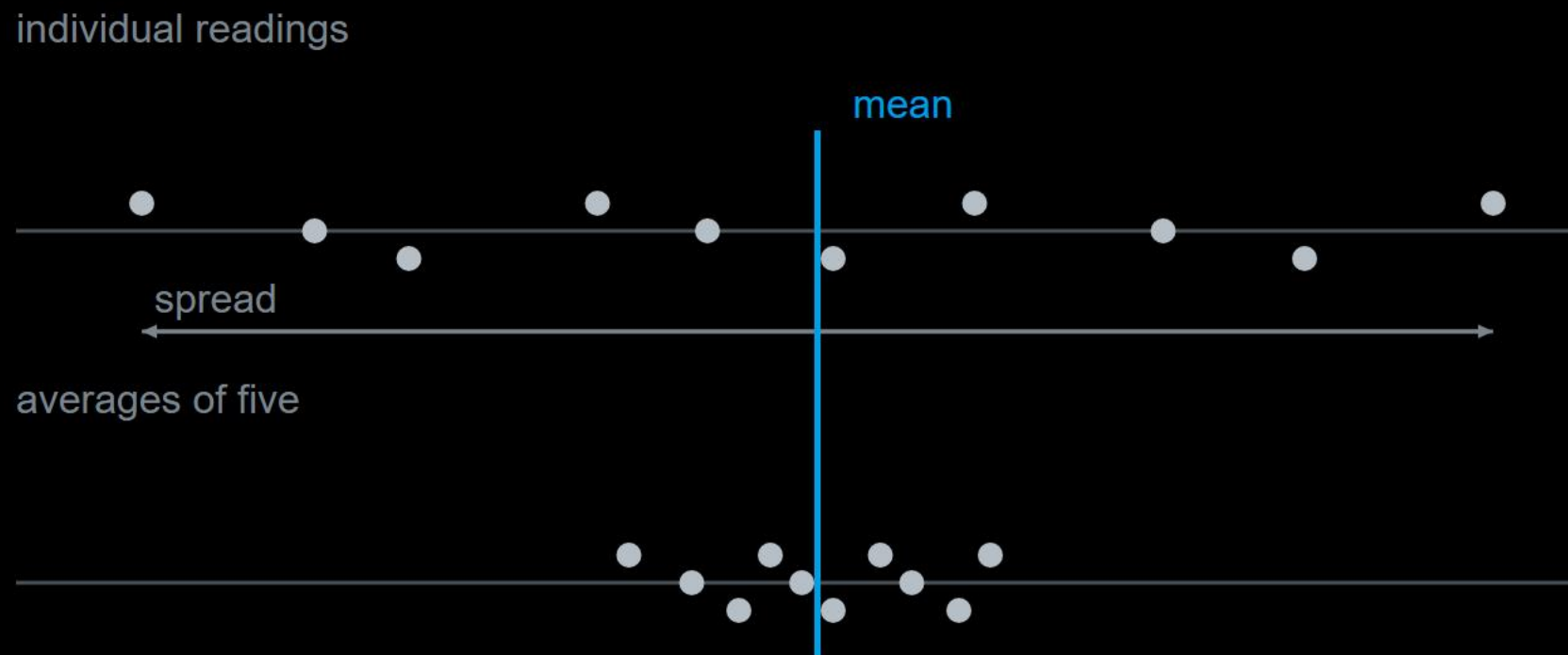
individual readings



averages of five



a series reveals what one reading hides



report the centre and the spread.

the knob: integration time

few long readings



one second

one reading is a guess

the knob: integration time

few long readings



more, shorter readings



one second

the knob: integration time

few long readings



more, shorter readings



the band that still
reads correctly

many short readings



these land in the
wrong colour

one second

one reading is a guess

longer is calmer.
shorter is faster.

where is your limit? measure it.

part 4

build evidence

two teams improve their result

team a

changed distance and
integration time

recognition improved

team b

changed only distance

recognition improved

two teams improve their result

team a

changed distance and
integration time

recognition improved



what caused it?

team b

changed only distance

recognition improved



distance

change one variable at a time.

what makes a measurement series

before

write conditions
write the expectation

what makes a measurement series

before

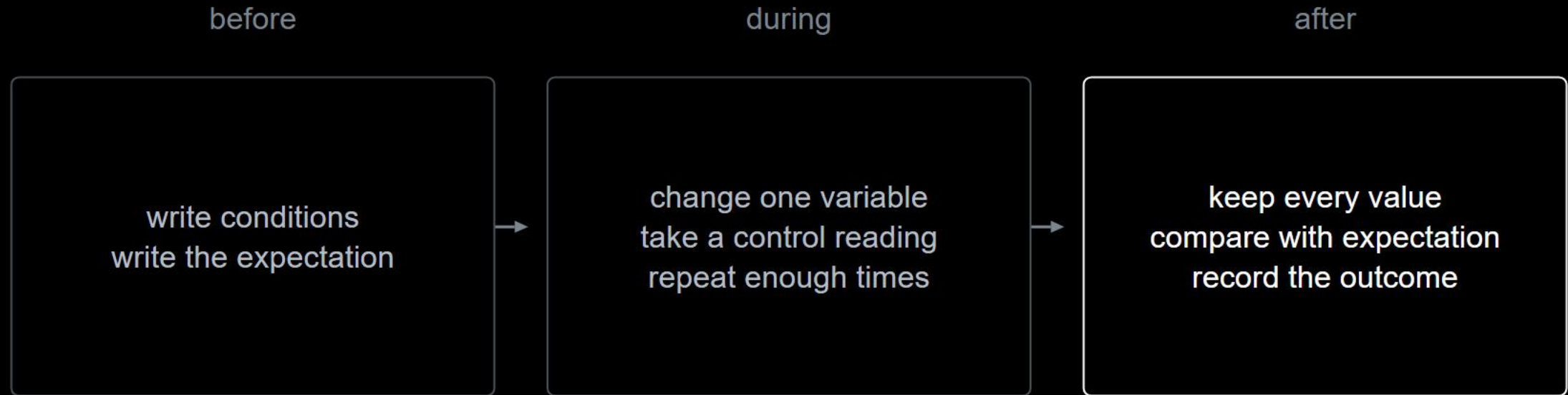
write conditions
write the expectation



during

change one variable
take a control reading
repeat enough times

what makes a measurement series



repeatable beats impressive.

luck or skill?

5 / 5

50 / 50

guessing succeeds about once in 32 attempts

luck or skill?

5 / 5

guessing succeeds about once in 32 attempts

50 / 50

guessing: about 1 in 1,125,899,906,842,624

luck or skill?

5 / 5



guessing succeeds about once in 32 attempts

50 / 50



guessing: about 1 in 1,125,899,906,842,624

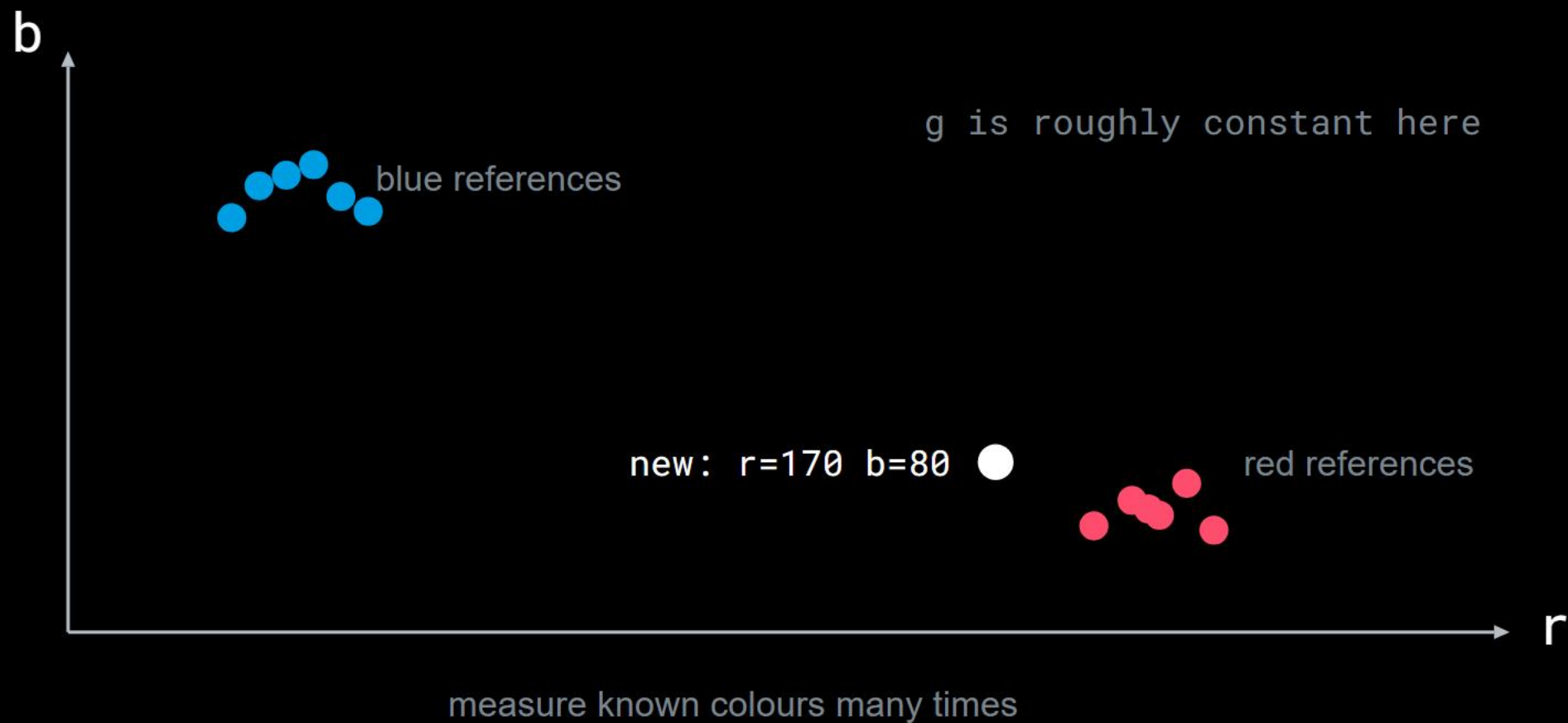
evidence

calibrate, then classify

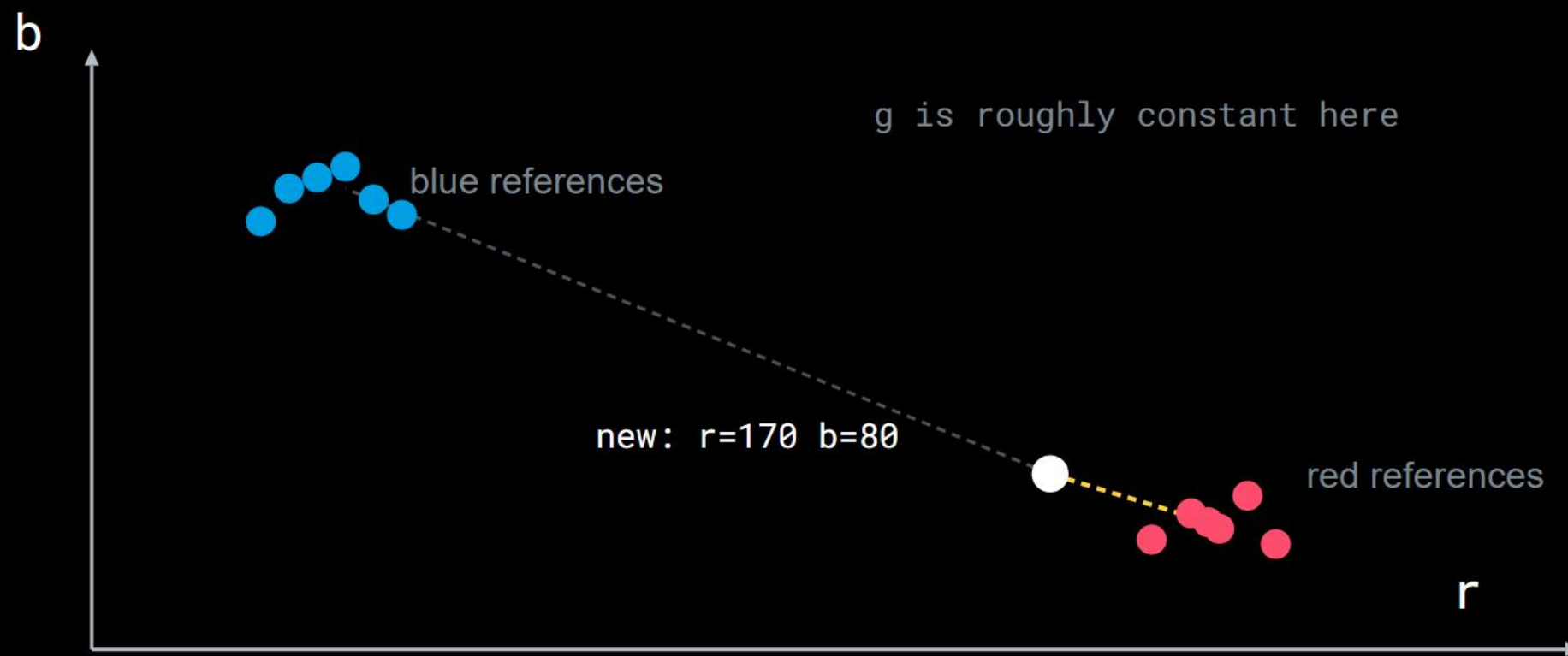


measure known colours many times

calibrate, then classify

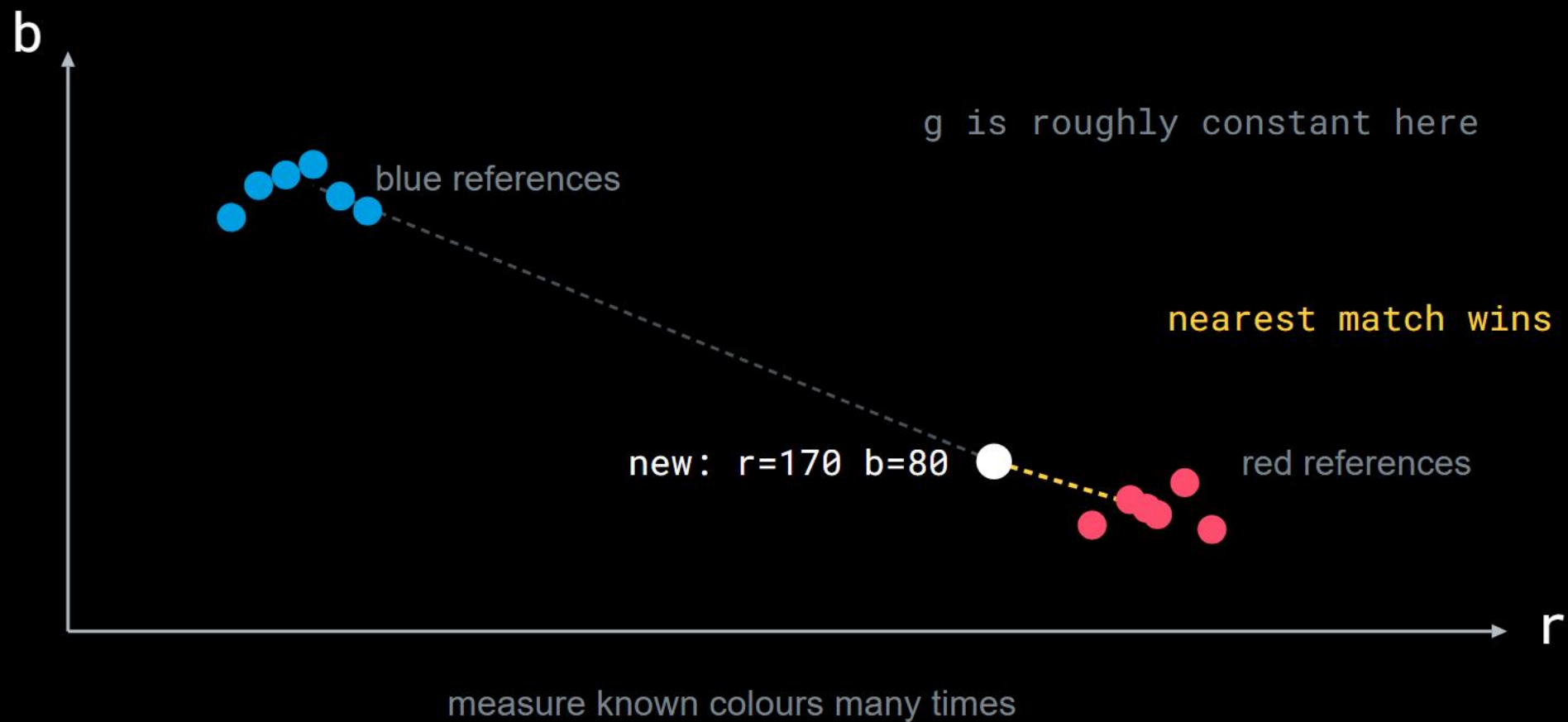


calibrate, then classify



measure known colours many times

calibrate, then classify



when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"

your measurement series

gain ↑
recognition: 94% → 82%

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"

your measurement series

gain ↑
recognition: 94% → 82%

✓ one variable

✓ enough repetitions

✓ control reading

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"



your measurement series

gain ↑
recognition: 94% → 82%



✓ one variable

✓ enough repetitions

✓ control reading

ground truth for your setup

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"



your measurement series

gain ↑
recognition: 94% → 82%



✓ one variable

✓ enough repetitions

✓ control reading

ground truth for your setup

error log

suggestion



measurement



outcome

if you did not measure it,
you do not know it.

one reading is not a measurement series.